Chaturanga and the Evolution of Chess

A research report on the ancient Indian game of the "four limbs," its stories, its oldest surviving pieces, and its long journey to modern chess

---

How to read this report

This report follows one idea across about fifteen centuries: a battlefield turned into a board. It begins with the Sanskrit word *chaturanga* and the army it described, moves through the earliest written and archaeological evidence, retells the legends people told about the game (including the ones connected with the Mahabharata and Ramayana), explains how the game was played and why, and then follows it through Persia, the Arab world, East Asia and Europe until it becomes the chess played today.

One rule runs through the whole report: **separate what is documented from what is legend.** Chess has more beautiful origin stories than almost any other human invention, and most of them are not history. Where a story is a legend, a later invention or a modern novel, the text says so. Where scholars disagree, both sides are given.

The main scholarly foundation is H. J. R. Murray's *A History of Chess* (1913), later refined by Henry Davidson, Harry Golombek, Richard Eales and others, together with Persian, Arabic and Sanskrit sources and archaeological finds from Central Asia. Sources are listed at the end.

Summary in one page

- **Name.** *Chaturanga* (Sanskrit *caturaṅga*) means "having four limbs" or "four parts." In epic poetry it commonly means "army." The four parts were elephants, chariots, cavalry and infantry.
- **Where and when.** The prevailing view is that the game arose in northern India in or before the early seventh century CE, in the later Gupta period (the sixth century is the usual estimate). The first clear mention of the name in connection with the game comes from Banabhatta's *Harshacharita*, written around 625 CE.
- **What it was.** A two-player war game on an 8×8 uncolored board called the *ashtapada*. Each side had a king (raja), a counsellor (mantri), two elephants (gaja), two horses (ashva), two chariots (ratha) and eight foot soldiers (padati). The exact rules were

never recorded in a surviving Indian manual from that early period, so they are reconstructed from its descendants.

- **How it spread.** To Sasanian Persia (where it became *chatrang*, around the reign of Khosrow I, 531–579), then to the Arab world (*shatranj*), then through Muslim Spain and Italy into Europe. Eastwards it became the Southeast Asian games (Burmese *sittuyin*, Thai *makruk*, Cambodian *ouk chatrang*), and probably the Chinese *xiangqi*, Korean *janggi* and Japanese *shogi*.
- **Oldest pieces.** The oldest securely identified chess pieces are the seven ivory figures from Afrasiab (ancient Samarkand, Uzbekistan), dated to roughly the seventh to eighth century. Older figures from Dalverzin-tepe are debated.
- **Modern chess.** The powerful queen and bishop appeared in Spain and Italy around 1475 (the poem *Scachs d'amor*, Valencia). The modern tournament world began with London 1851, the Staunton pieces (1849) and FIDE (1924).
- **Legends.** The Mahabharata contains a famous dice game but not chess. Chess legends attach to Mandodari (wife of Ravana), to the Persian sage Bozorgmehr, to the Indian wise man Sissa and to Shiva and Parvati. None is proven history.

Timeline at a glance

Period	Event	Status
c. 1500–500 BCE	Dice and board games exist in India; the word *chaturanga* is used for armies	Documented (the word), not chess
c. 5th century BCE onward	Buddhist texts list board games on 8 and 10 rows (*ashtapada*, *dasapada*) that monks should avoid	Documented, but these are not clearly chess
c. 300–500 CE	Mahabharata reaches its final form; it describes the four-part army and the dice game	Documented
c. 5th century CE	Chinese pilgrim Fa Xian travels India and does not mention chess	Argument from silence
c. 600 CE	Subandhu's *Vasavadatta* uses a chessboard image (frogs on black squares)	Documented, interpretation debated

Period	Event	Status
c. 600–625 CE	*Harshacharita* by Banabhatta names *chaturanga* alongside the ashtapada	Earliest clear Sanskrit mention
6th–7th century	Pahlavi (Middle Persian) texts describe *chatrang* arriving from India	Documented, with legendary content
c. 7th–8th century	Afrasiab ivory set, Samarkand	Oldest securely identified pieces
8th–9th century	Abstract pieces at Nishapur; Arabic writers on *shatranj*	Documented
c. 1000	Ferdowsi's *Shahnameh*; al-Biruni describes four-handed *chaturaji*	Documented
11th–12th century	Chess spreads across Europe; the Charlemagne and Lewis pieces are carved	Documented
c. 1150	*Manasollasa* in India describes chaturanga among royal pastimes	Documented
c. 1475	Queen and bishop gain modern powers (*Scachs d'amor*, Valencia)	Documented
1849–1851	Staunton pieces; first international tournament in London	Documented
1886	First official world champion, Wilhelm Steinitz	Documented
1924	FIDE founded in Paris	Documented
1988	Viswanathan Anand becomes India's first grandmaster	Documented

Chapter 1. The word and the idea: "four limbs"

1.1 What *chaturanga* means

Chaturanga is a Sanskrit compound. *Chatur* means "four" and *anga* means "limb," "part" or "member." Grammarians call a compound of this type a *bahuvrihi*: it names a thing by describing what it possesses. A *chaturanga* is therefore "that which has four limbs."

In the ancient Sanskrit epics the word is used again and again for an **army with four arms**:

1. **Elephants** (*gaja*, *hasti*), the armored heavy units, the tanks of the ancient world;
2. **Chariots** (*ratha*), the mobile platforms for the great warriors and their archers;
3. **Cavalry** (*ashva*, horse), the fast scouts and flankers;
4. **Infantry** (*padati*, *pada*, *bhata*), the foot soldiers who made up the bulk of any host.

This fourfold division of military strength was an established way of thinking in Indian statecraft long before any board game used it. When a poet in the epics wrote that a king marched out with a *chaturanga* force, the audience understood an organized army with all four branches present.

1.2 The army in the Mahabharata

The Mahabharata gives the fourfold army a precise mathematical structure. Its basic unit is the *akshauhini*, a full army division. According to the epic's counting scheme, an akshauhini is made up of:

- 21,870 chariots,
- 21,870 elephants,
- 65,610 horses, and
- 109,350 foot soldiers.

Notice the ratios. For every chariot there is one elephant, three horses and five foot soldiers. The Kurukshetra war in the epic is said to involve eighteen akshauhinis in total, eleven on the Kaurava side and seven on the Pandava side. Whatever one thinks of the figures (they are literary, not census numbers), they show how systematically the epic thinks about armies as four-part structures with fixed proportions.

Some chess historians have pointed out that the akshauhini's arrangement is reminiscent of the starting setup of chaturanga, with chariots, elephants and horses represented in pairs and infantry in the front rank. The comparison is suggestive, but it should be treated with care: the epic describes battle arrays, not board games, and the resemblance may be partly the result of the game having been designed to imitate the army as Indians understood it.

1.3 From army to game

The key conceptual step is simple but enormously important. Somebody, at some point, realized that a war could be modeled on a grid. Each unit type could be given a distinct way of moving that expressed its character:

- The **elephant** is heavy, powerful, hard to stop.
- The **horse** leaps over obstacles and strikes from unexpected angles.
- The **chariot** charges in straight lines at speed.
- The **foot soldier** advances slowly, one step at a time, and captures only what is in front of him diagonally, since spears and swords reach forward.
- The **king**, who must not fall, moves one careful step at a time.
- The **counsellor** (*mantri*, minister) stays close to the king and moves only a short distance.

The winning condition was equally logical. A real battle was lost when the king was captured, so the game was decided by the king's fate. That one decision, to make victory depend on a single piece, is what makes chaturanga the ancestor of all later chess. Encyclopaedia Britannica singles out exactly these features: pieces with different powers, and a win decided by the fate of one piece.

That is why chess, even today, is described as a war game, with its "attacks," "defenses," "sacrifices" and "campaigns." The metaphor is not a modern flourish. It is the founding idea.

1.4 How the name travelled

The Sanskrit word did not stay in India. It was borrowed and reshaped by every culture that received the game:

- **Persian (Middle Persian):** *chatrang*. The vowels were worn down, a natural process when a foreign word is borrowed.

- **Arabic:** *shatranj* (the Arabs have no "ch" sound in this position and no "ng" cluster at the end, so the form changed again).
- **Spanish and Portuguese:** *ajedrez*, *xadrez*, from Arabic *ash-shatranj*.
- **Greek:** *zatrikion*, taken from Persian *chatrang*.
- **Tibetan:** *chandaraki*, plainly derived from *chaturanga*.
- **Burmese:** *sittuyin*. In Burmese *sit* means "army," so the name keeps the sense of the original.
- **Malay/Indonesian:** *catur*, from *chatur*.
- **Thai:** *makruk* (a different word, but the game itself belongs to the same family).
- **Cambodian:** *ouk chatrang*, again plainly from *chaturanga*.

The survival of the Sanskrit root across so many languages is itself one of the strongest pieces of evidence for the Indian origin of the game. When a name travels with a game, it usually tells you where the game came from.

Chapter 2. Before chess: boards, dice and the *ashtapada*

2.1 The ashtapada board

Ancient Indian literature mentions a board called the **ashtapada**, which in Sanskrit means "eight-footed" or "eight-stepped." It is a square board of 8×8 squares, 64 in all, and in Indian practice it was **not** colored in alternating light and dark squares. This is a detail that surprises modern players: the checkered pattern of today's boards came much later. Early chaturanga boards were plain.

Some ashtapada boards carry special markings on certain squares. The purpose of these marks is not known. Murray suggested that the board may also have been used for a race game played with dice, in which such marks had a function, and that the marks were simply kept on the board by tradition when the new war game borrowed it. Others note that the marked squares coincide with positions that certain pieces could never reach under the reconstructed movement rules, which supports the idea that the marks belonged to some earlier use.

The important point is that the **board came first**. A grid of 64 squares existed as a gaming surface before chess did. When the army game was invented, its designers placed it on the familiar ashtapada, and the game took on the name *chaturanga*, the "army game." One

careful chess historian summarizes it this way: when chess was put on the ashtapada, it was natural to call it chaturanga.

2.2 Dice games in ancient India

India had a very old and very powerful tradition of dice. Dice appear in the Rigveda, and gambling with dice is one of the most dramatic episodes in the Mahabharata (see Chapter 4). A whole class of Indian board games, the race games, combined board and dice. The familiar *pachisi* (the ancestor of Ludo and Parcheesi) and *chowka bhara* belong to this tradition.

This matters for chess history because several early theories assumed that chaturanga **was itself a dice game**. We will return to this under the Cox–Forbes theory below. The mainstream view today is the opposite: two-player chaturanga, as the ancestor of chess, was a game of pure skill, and the dice-driven four-player variant (*chaturaji*) is a later offshoot.

2.3 Buddhist lists of games

Early Buddhist texts, such as the Brahmajala Sutta and the monastic rules (the Vinaya), contain lists of entertainments and games that monks should not indulge in. These lists include board games played on boards of eight rows (*ashtapada*) and ten rows (*dasapada*), as well as games played on an imaginary board in the air. Some earlier scholars, including Murray in his first treatment, tried to read these passages as evidence for early chess. Murray later became unsure about that interpretation, and most specialists today treat the passages as evidence for **board games on 8×8 and 10×10 grids**, not chess in particular.

Still, the lists are valuable. They show that gridded game boards were common and socially noticed in the centuries around the time of the Buddha, and that religious teachers saw such games as distractions. That attitude toward board games, as time-wasting or morally risky, would follow chess for the next two thousand years.

2.4 The Cox–Forbes theory

In the late eighteenth century a British officer in Burma, Hiram Cox, came across the four-handed dice game later called **chaturaji** ("four kings"). In the nineteenth century Duncan Forbes developed the idea into the **Cox–Forbes theory**: that four-handed, dice-driven

chaturaji was the original form of the game, and that two-player chess was a later simplification created by removing the dice and merging the allied players.

The theory had an attractive logic. The name *chaturanga* means "four limbs," and a game with four players, each with an army, seemed to match it nicely. A tradition (reported in medieval and later sources) also held that the dice were removed on religious grounds, since gambling was forbidden.

Murray's 1913 study dismantled it. His main arguments were:

1. The earliest detailed description of the four-handed game is by al-Biruni around 1030, **centuries after** the first mentions of the two-player game in Persian and Sanskrit.
2. The Persian texts about chess describe a two-player game with no dice.
3. Words such as *chaturanga* were used for "army" long before any game, so the name does not by itself imply four players.

Today most scholars regard chaturaji as a **derivative** of chaturanga, an adaptation that added dice and extra players. Still, the Cox–Forbes theory is historically important: it shows how seriously nineteenth-century scholars took the Indian roots of the game, and it explains why you will still find it in older books and popular articles.

2.5 The Lothal claim

A further claim sometimes appears in popular Indian writing: that archaeologists found chaturanga pieces at the Indus Valley site of Lothal, dated by radiocarbon to about 2500 BCE, proving that the game is 4,500 years old. This claim has been challenged by chess historians. The objections are that no proof links the objects to chess, that the 64-square board in early India belonged to a dice race game, and that no written evidence for chess exists before about the sixth century CE. In the debate over the claim, one participant described it as reviving the old Cox–Forbes theory, which Murray had already shown to be unsupported.

The responsible summary is this: the Indus civilization certainly had games, dice and gaming pieces, but there is **no accepted evidence that it had chess**.

Chapter 3. The first written evidence

3.1 An argument from silence

Historians of chess have long noticed how quietly the game enters history. Murray and Richard Eales concluded that before the seventh century, the existence of chess in any land cannot be demonstrated. One often-cited observation is that the Chinese Buddhist pilgrim **Fa Xian**, who travelled widely through India at the beginning of the fifth century and wrote a detailed account, never mentions chess. An argument from silence is never conclusive, but taken with the lack of early references it suggests that the game did not yet exist, or was not yet widely known, around 400 CE.

Then, within a few decades around the year 600, references suddenly appear in several languages. That cluster is why scholars place the invention around the sixth century.

3.2 Subandhu's frogs (c. 600)

The Sanskrit prose romance *Vasavadatta*, by Subandhu, describes the rainy season in a playful image: the season plays a game, with frogs as the playing pieces, yellow and green in color, as though stained with lac, leaping onto the black squares of the board. The passage appears to describe a chessboard-like game with two-colored pieces. Some scholars read it as an early literary allusion to chess; the poem does not say the word *chaturanga*, so it is not conclusive.

3.3 Banabhatta's *Harshacharita* (c. 625)

The most important Sanskrit reference is in the *Harshacharita*, the biography of the emperor **Harsha** (who ruled from 606 to 647) by his court poet **Banabhatta**. In a passage praising the king's peaceful rule, the poet uses a series of puns: under this king only bees quarrel (over the dew), the only feet that are cut off are the feet of verse (in prosody), and **it is only on the ashtapada that one learns how to draw up a chaturanga**, since no one's four limbs are cut off in real punishment.

There are two things to understand here. First, the pun on *chaturanga* works because the word has two meanings: the four-part army, and the game. Second, the same line mentions the *ashtapada*, the board, so the poet associates the game directly with its 64-square board. This is generally accepted as the **earliest clear reference to the name**

chaturanga in connection with the game. Scholars note that the passage also gives the first known use of the idea that the game teaches the arrangement of an army without the cruelty of war, a theme that would be repeated for centuries.

3.4 The Persian witnesses

Three Middle Persian (Pahlavi) texts mention *chatrang*:

1. **The Karnamak-i Ardashir-i Papakan**, a legendary account of Ardashir I, founder of the Sasanian dynasty (r. 224–241), which lists chess among his accomplishments. Because the work is full of legends and was written much later than Ardashir's lifetime, it proves only that chess was popular when the text was composed.
2. **Khusraw ud Redag** ("Khosrow and the Page"), in which a boy boasts of his skill in chess, backgammon and other arts. This text also seems to be late.
3. **Wizarishn i Chatrang** ("The Explanation of Chess"), also known as the *Chatrang-namak*, which tells the most famous story of how chess arrived in Persia. It is treated in detail in Chapter 8.

The Pahlavi texts do not date themselves, and the first two may include post-Sasanian editing, so they cannot independently prove an early date. The *Wizarishn* is the key text, and its date (around the early seventh century) is consistent with the Sanskrit evidence.

3.5 What the earliest sources agree on

Putting the earliest sources side by side, certain points are consistent:

- The game is **Indian in name**, because every early Persian and Arabic name is derived from the Sanskrit.
- The game is **military in character**, a model of an army.
- It was a **courtly, aristocratic pastime** from the beginning, associated with kings, ministers and ambassadors.
- It was already so sophisticated that it served as a **test of wisdom** between kingdoms, as the story of the Indian envoy at Khosrow's court shows.

What the earliest sources do **not** give is a complete rulebook. Murray observed that because Sanskrit writing on chaturanga is so sparse, the original game may be lost to history, and that knowledge of it comes second-hand, mostly from Muslim sources describing shatranj. This is an essential caution for the next chapters.

Chapter 4. Stories of chess: the Mahabharata, the Ramayana and the legends

The first thing a student must know about "chess in the Mahabharata" is that the question has a surprising answer. **The Mahabharata does not describe a game of chess being played.** What it contains is something else: the most famous gambling scene in world literature, a single passage in which a board-and-dice game is mentioned, and an enormous amount of thinking about armies and battle formations. Together these made the soil in which chaturanga grew. This chapter separates the pieces.

4.1 The great dice game (Sabha Parva)

In the Sabha Parva, the book of the assembly hall, the Kaurava prince **Duryodhana** returns from the Pandavas' royal consecration (the Rajasuya sacrifice) consumed by jealousy of his cousin **Yudhishthira**. His uncle **Shakuni**, the prince of Gandhara and the brother of Gandhari, is a master of dice and, according to the tradition, a cheat. Shakuni proposes a game of dice. Yudhishthira, bound by his code of honor not to refuse a challenge, accepts.

What follows is the turning point of the whole epic. Yudhishthira loses his wealth, his kingdom, his brothers and finally himself. He then stakes and loses **Draupadi**, the common wife of the five Pandavas, who is dragged into the hall and humiliated. The result of the game is the exile of the Pandavas for thirteen years, and the grievance that leads to the Kurukshetra war.

Two points are worth making for chess history:

1. This was a **game of chance**, not skill. The tragedy of the scene is that Yudhishthira has no way to out-think a loaded set of dice.
2. The episode helps explain why, for centuries afterwards, Indian moral and religious writers warned against gambling, and why a skill-only game like chaturanga could later be presented as a respectable alternative.

The relief carving of Shakuni and Yudhishthira playing dice at Lepakshi (in Andhra Pradesh) is one of the best-known images of this scene in Indian art. It shows a game, but not chess.

4.2 Yudhishthira as Kanka (Virata Parva)

The passage that has made some readers think of chess comes in Book 4, the Virata Parva. During the thirteenth year of exile the Pandavas must live in disguise. Yudhishthira explains to his brothers that he will present himself to King Virata as a Brahmin named **Kanka**, a courtier skilled in dice. In Kisari Mohan Ganguli's 1896 English translation he says that he will entertain the king and his friends by moving on **boards beautiful pieces of ivory, in blue, yellow, red and white colors, by throws of black and red dice**.

This is the passage popularly cited as "chess in the Mahabharata." It deserves careful reading:

- The pieces are made of **ivory** and come in **four colors**, which would suit a four-player game.
- They are moved by **throws of dice**, so luck decides what is played.
- It occurs in the Virata Parva, a section whose date is not precisely known.

Scholars read it in different ways. One view is that it points to a four-handed, dice-driven game like **chaturaji**. Another is that it describes a race game such as **pachisi**. A third suggestion is that it is an early form of a chess-like game. What it very probably does **not** describe is the two-player, no-dice chaturanga of the sixth century. The honest conclusion is that the Mahabharata knew board games with pieces and dice, and that such games formed part of the background from which chaturanga emerged.

A point of dating is useful here. The Mahabharata is believed to have reached something like its final form by the early Gupta period, around the fourth century CE, but parts are much older and parts may be later. A passage cannot therefore be used to date chess to the time of the war itself, which tradition places in a distant age.

4.3 The war as a living chessboard: battle formations

While the epic does not show chess being played, it shows armies being **arranged like pieces on a board**. The Kurukshetra war is full of named formations (*vyuhas*) that the commanders deploy each day:

- the **Chakravyuha** (also *Padmavyuha*), the circular lotus formation, inside which the young hero **Abhimanyu** is trapped and killed after he enters knowing how to break in but not how to break out;
- the **Garuda vyuha** (eagle), the **Kraunca vyuha** (heron), the **Makara vyuha** (crocodile) and others.

The point is that the epic's heroes and poets thought of the battlefield geometrically. Commanders positioned elephants, chariots, cavalry and infantry in relation to one another, and those who understood the patterns won. This habit of mind, an army as an arrangement of different units in space, is exactly the idea that a board game turns into rules.

A caution is needed. Modern writers love to say "Kurukshetra was a chessboard" and to call Krishna a grandmaster who moved the Pandavas like pieces. That is a lovely **metaphor**, and it is not evidence that chess existed in the epic age.

4.4 Krishna, Radha and the game

In later Indian painting, particularly from the Rajput and Pahari schools of the medieval and early modern period, one finds pictures of the divine couple **Krishna and Radha** seated at a chessboard. The painting called "Krishna and Radha Playing Chaturanga," now widely reproduced, shows the pair on an 8×8 board. These images belong to the tradition of devotional love-play (*lila*) and show that chess had become, by that period, a graceful courtly amusement associated with romance. They are not ancient evidence, but they are valuable pictures of how the game was imagined in India. A popular school textbook in Rajasthan also states that Krishna and his friends are believed to have played the game in the Mahabharata age; this belongs to tradition, not to the text of the epic.

4.5 Mandodari and the invention of the game (Ramayana tradition)

The most popular Indian origin story places the invention of chess in **Lanka**, in the Ramayana world. It is told like this.

Ravana, the ten-headed king of Lanka, is a warrior who knows only conflict. His queen **Mandodari** is distressed by his constant wars and rage. She wants to turn his attention from real battle to something else. She invents a game of war on a board, with pawns, elephants, soldiers, kings and plenty of strategy, so that her husband can fight his battles on a board and not in the world. According to the story, she soon begins winning against him. The war of the Ramayana happens regardless.

A version of the legend adds divine help: Mandodari prays to **Ganesha**, who teaches Ravana the game and asks him to treat it as a real battlefield. Another version in the same tradition says that the daughter-in-law of Ravana, the wife of his son Meghnada, also took up the game. These stories are repeated in popular magazines, newspaper features and a school textbook for Rajasthan's Class 10 physical education course.

How should a careful reader treat this? Two points:

- **No ancient text is known to contain it.** I did not find a Sanskrit or Persian source for the Mandodari story in the research done for this report. It appears to be a later popular tradition. The Ramayana itself, in its standard versions, is silent about chess.
- **It has become literature in its own right.** A 2014 novel, *The Queen's Play* by Aashish Kaul, retells the story, imagining that the original game was played by four players with dice and that Mandodari invented the two-player version by discarding the dice and promoting the queen figure. That is imaginative fiction, not history.

What is interesting about the legend is its **meaning**. It tells us that Indians understood chess as a *substitute for war*: a way of draining violence into a board. It also explains the theme that the game was invented by a woman for a restless king, a theme that appears again in Persian and Arabic stories where a sage invents the game to teach a king a lesson.

4.6 Shiva and the Tamil chess legend

South Indian temple tradition has its own chess story. In Tamil Nadu, the god **Shiva** is said to have played chess disguised as a *Siddha* (a perfected ascetic). He won the game, threw off his disguise and revealed himself to everyone; this was followed by a grand wedding with the goddess **Rajarajeswari**. In memory of the victory, the Lord is worshipped under the name **Chaturanga Vallabha Nathar**, "Lord who is the master of chaturanga," from the Tamil form *chaturangam* for the game. The legend is attached to a temple tradition and shows that chess had entered the religious imagination of Tamil culture, where, in later times, the game was a popular pastime.

4.7 The Indian wise man and the Persian king (Sissa)

The best-known **Arabic and Persian** origin story concerns a wise man called **Sissa ibn Dahir** and a king, whose name varies (Balhait, Shahram, Shihram, Shirham). It ends with the famous wheat-on-the-chessboard reward and is told in Chapter 14, because it is as much a story about mathematics as about chess. An older form of the same legend tells of a queen named Husiya, daughter of Balhait, whose son was killed by a rebel. The news was gently given to the grieving mother by showing her, on the chessboard, how the prince had fallen. The first known writer to record the wheat story is the thirteenth-century biographer **Ibn Khallikan** (1256).

4.8 Why the legends matter

The legends differ in names and places, but they share a pattern:

Legend	Inventor	Purpose of the invention
Mandodari	A queen	Distract a warlike king from real war
Ganesha and Ravana	A god	Teach strategy as a substitute for battle
Sissa	A wise man	Teach a king to value his subjects (and show the value of the lowly pieces)
Husiya's story	An Indian sage	Break news of a death gently through the game
Bozorgmehr (Persia)	A minister	Win a diplomatic contest of wits

In every case chess is a **substitute for violence** or a **teacher of wisdom**. That is how the cultures that played it understood it.

Chapter 5. How chaturanga was played

5.1 A warning about the rules

No complete Sanskrit rulebook for chaturanga survives from the early period. What we know comes from comparing:

1. the Persian *chatrang* described in Pahlavi texts;
2. the Arabic *shatranj* of the ninth and tenth centuries, which is well documented in chess manuals;
3. the rules of the Asian descendants (Burmese, Thai, Chinese and others);
4. the later Indian descriptions of the game (for example in the *Manasollasa*).

Murray, Davidson, Golombek and other historians have each written a reconstruction. They agree on most points, and differ on a few. What follows is the mainstream picture, with the disputes marked.

5.2 The board and the setup

- **Board:** 8×8 squares (the *ashtapada*), a single color. Alternating colors came much later.
- **Players:** two, one against the other.
- **Pieces:** sixteen per side, eight in the back rank and eight foot soldiers in the second rank. They were originally **figurative**: carved elephants, horses, chariots and soldiers. The Islamic world later made them abstract, probably out of religious reluctance to represent living creatures.
- **Back rank:** chariot, horse, elephant, then the royal pair (king and minister), and the same three on the other side, matching the modern setup of rook, knight, bishop and king/queen.

One point about the setup in the Arabic game is worth noting: the position of the two kings was not fixed in shatranj, so a player could place the king on either side. Whether this was also true of chaturanga is not known.

5.3 The pieces

Chaturanga piece	Meaning	Reconstructed move	Modern descendant
Raja (king)	The ruler	One square in any direction	King
Mantri (minister, also *senapati*, general)	The counsellor	One square diagonally	Queen (much weaker)
Gaja (elephant, also *hasti*)	Elephant corps	Disputed; in shatranj, two squares diagonally, leaping	Bishop
Ashva (horse)	Cavalry	Two squares in one direction, then one square at a right angle, leaping	Knight
Ratha (chariot, also *nauka* boat in some versions)	Chariot corps	Any distance along ranks and files	Rook
Padati (foot soldier, also *bhata*, *sainika*)	Infantry	One square forward; captures one square diagonally forward	Pawn

A few observations:

- The **mantri** is a modest piece. The overwhelming power of the modern queen is a European creation of about 1475.
- The **gaja** is the greatest mystery. Murray argued that it moved exactly two squares diagonally and could jump over an intervening piece, as the *alfil* does in shatranj. Others have proposed different moves, such as one square forward or diagonally (like the legs and trunk of an elephant, resembling the silver general of Japanese shogi), or two squares along ranks and files. The varied moves of the elephant in the Asian descendants (the Thai and Burmese elephant resembles the silver-general move, the Chinese elephant is a two-square diagonal piece that cannot jump) show how the move may have changed as the game spread.
- The **ratha** has also been debated. The German historian Johannes Kohtz suggested that the original chariot move may have been different from the rook's, but most scholars accept the straight-line move.
- The **padati** had no two-square first move. Pawns advanced one square at a time, which made opening play slow.

5.4 The rules of play

The following rules are those of shatranj, which historians assume closely resembled chaturanga:

- **No castling.** It was invented much later in Europe.
- **No en passant**, since there was no double step.
- **Pawn promotion** was to the weak minister (mantri) only. In some descriptions a pawn could promote to the piece originally on its file; the early rule is uncertain.
- **Check** was announced, and a king in check had to be saved. A king could not be left in check.
- **Win by checkmate** (the king is attacked and cannot escape).
- **Win by baring the king:** capturing every opposing piece except the king also won, though this rule may have been added to shatranj and was often countered by a draw if the opponent could bare your king on the next move.
- **Stalemate counted as a win** for the player who delivered it in shatranj, a rule that persisted for centuries. The modern rule that stalemate is a draw was fixed only in the early nineteenth century.

The Arabic master **al-Adli** (ninth century) records some further customs in the game, and the later literature gives detailed opening play, endgame studies and chess problems, but those belong to the Islamic stage described in Chapter 9.

5.5 How a game felt

Players of reconstructed shatranj or chaturanga report a game very different from modern chess:

- **Slow and positional.** With weak pieces (the minister and elephant) and slow pawns, attacks took time to build, and a lead in material was often decisive.
- **Tactics matter less than patience.** There is no long-range queen to launch sudden raids.
- **Endgames were central.** Because the king had to be mated or the opponent bared, endgame technique was a major part of the art.

The literature of the Islamic masters shows that players of the old game developed openings (*tabiyat*), problem positions (*mansubat*) and strategy that already resemble what modern players study.

5.6 The four-handed game: Chaturaji

The four-player variant, **chaturaji** ("four kings"), was described in detail by **al-Biruni** in his book on India (about 1030). It is the version that fits the Mahabharata's Kanka passage best:

- **Four players**, each with a color (often black, green, red and yellow) and a set of pieces: a king, a horse, a boat (*nauka*), an elephant and four pawns.
- Players sit at the four sides of the board, and two sets of players are allies.
- **Dice decide the move.** Two dice were thrown; the number rolled determined which piece could be moved. In one reconstruction with a six-sided die, a 1 or 2 moved the king or a pawn, a 3 the boat, a 4 the horse and a 5 the elephant, while a 6 allowed no move. Some descriptions use four-sided stick dice with a different assignment. Murray believed cube dice were a substitute for older oblong dice.
- The **boat** moves two squares diagonally, the same move that Murray assigned to the elephant of shatranj.
- The term *chaturaji* itself was also used for a position similar to checkmate (a king reduced to isolation).

As noted in Chapter 2, scholars now regard chaturaji as a dice-driven offshoot of two-player chaturanga, not its ancestor. Still, it kept the memory of the Indian game alive in Indian homes into the twentieth century.

5.7 What changed between chaturanga and modern chess

Feature	Chaturanga / shatranj	Modern chess
Board colors	One color	Alternating light and dark
Minister/queen	One square diagonally	Any distance in any direction
Elephant/bishop	Short diagonal leap or other short move	Any distance diagonally
Pawn first move	One square only	One or two squares
En passant	None	Yes
Castling	None	Yes
Promotion	To the weak minister	To queen, rook, bishop or knight
Stalemate	A win for the stalemating player (shatranj)	A draw
Bare king	A win	Not a win
Pace	Slow, material-driven	Fast and tactical

Chapter 6. Why they played: the purposes of the game

Chaturanga was never "just a game." In every culture that received it, people gave reasons for playing, and the reasons reveal what the game meant. The sources point to several overlapping purposes.

6.1 A model of war without bloodshed

The earliest Sanskrit reference, in Banabhatta's *Harshacharita*, already presents the game as a **peaceful stand-in for war**: under a good king, one learns how to arrange a chaturanga only on the board, not by cutting off real limbs. Historians have called this the first known description of chess as a substitute for, or a training in, actual war.

It is important not to overstate this. There is no firm evidence that Indian armies used chaturanga as a formal drill. What the evidence shows is that the game was **designed as a model of an army** and **understood by players as a battle in miniature**. A king who played could rehearse the thinking that war requires: protecting the ruler, coordinating different kinds of troops, sacrificing material for position, and judging an opponent's plans.

6.2 Education of princes and nobles

A second purpose was **education**. The Persian text *Khusraw ud Redag* lists chess among the skills that a well-bred young man should master, together with other courtly arts. In India, the twelfth-century encyclopedia **Manasollasa** ("Delight of the Mind" or "Delight of the Spirit"), written for the Chalukya king **Someshvara III** (reigned about 1126 to 1138), treats chess as one of the accomplishments and pleasures of a ruler. Scholars have described that work as a "mirror for princes," a handbook for the king's education, and it has a section on chaturanga alongside sections on other games and diversions.

6.3 Diplomacy, prestige and wealth

Chess was a **tool of diplomacy**. The famous Persian story (see Chapter 8) presents the arrival of the game as a gift from an Indian king, accompanied by a challenge. In the Persian tradition, the set sent to Khosrow was made of precious materials: sixteen emerald pieces and sixteen ruby pieces. Whether or not the details are true, the story assumes that chess sets were fit gifts between kings.

The archaeological record confirms the point. Museum scholars studying early chess and backgammon pieces from Islamic lands have noted how many were made from **ivory and rock crystal**, expensive materials, which suggests that chess was an elite pastime. Historians also record that the Abbasid caliph Harun al-Rashid sent diplomatic gifts to Charlemagne's court at Aachen around the year 800, and that a famous medieval chess set later became attached to Charlemagne's name (see Chapter 7), even though the surviving pieces are much later.

6.4 A test of intelligence

In the Persian and Arabic stories, chess is repeatedly used as a **test of wisdom**. An Indian king sends the game to Persia to test the intelligence of the Iranians, and the Persian minister Bozorgmehr passes the test and answers with a game of his own. In another story the game's inventor shows a king how to be wise. In both cases the point is the same: a person who can understand a hidden structure of rules and strategy has real intelligence, and intelligence is power.

6.5 Moral teaching

A recurring lesson is that the game teaches **humility and care for the weak**. In the Sissa legend, the wise man designs a game in which the king, the most important piece, is helpless without the support of every other piece, even the lowliest pawn. The point of the story is to show a proud king that he depends on his subjects. This lesson, that the king is protected by the people he commands, is one of the most persistent themes in the cultural history of chess.

6.6 Courtship, leisure and the pleasures of court life

In India chess was also a courtly diversion associated with refined leisure. The Manasollasa describes games among the pleasures of the palace, and a recent scholarly study of another game in the same work, the race game *phanjika*, was titled "Gameplay as Foreplay at a Medieval Indian Court," a reminder that board games and courtship could go together. The later paintings of Krishna and Radha at a chessboard express the same idea: chess as a graceful setting for lovers.

6.7 Mathematical play: the knight's tour

Indian writers did not use the board only for competition. They were fascinated by **puzzles**, in particular the *knight's tour*, in which a knight visits every square once. A knight's tour appears in Sanskrit poetics (for example in connection with Rudrata's *Kavyalankara*, about 900 CE, and its commentaries) and again in the Manasollasa (about 1150). Later Indian authors worked out magic-square tours as well. The chessboard was a field of mathematical beauty as much as a battlefield.

6.8 Gambling, risk and religious unease

There was always a darker side. Because games had been tied to gambling (the Mahabharata's dice game is the great example), moral and religious authorities were suspicious of them. Buddhist texts list board games among the things monks should avoid. In the Islamic world jurists debated whether chess was lawful, with attention to two questions: whether it involved gambling and whether it distracted a person from prayer. In medieval Europe the Church criticized clergy who played. The attitude of suspicion followed the game wherever it went, and it did not stop people from playing.

Chapter 7. The oldest pieces and the oldest pictures

7.1 The surprising geography of the evidence

A reader might expect that the oldest chess pieces would be found in India, since that is where the game is believed to have been invented. They are not. The oldest securely identified chess pieces come from **Central Asia**, from the ruins of Afrasiab near Samarkand in what is now Uzbekistan. India has yielded very few early pieces. One partisan Iranian essay argues that this absence weakens the Indian-origin theory. Most scholars reply that the early Indian pieces may simply have been made of perishable materials (wood, clay) that did not survive, that the written and linguistic evidence points to India, and that the Silk Road explains why pieces appear in Central Asia soon after the game appears in Persia.

7.2 The Afrasiab chessmen (c. 7th–8th century)

- **Where found:** the settlement mound of Afrasiab, the ancient city beside modern Samarkand, Uzbekistan.
- **When found:** 1977, in excavations by the archaeologist **Yuri (Jurij) Buryakov**, as part of a large project of the Uzbek Academy of Archaeology.
- **What:** seven ivory figures: a **king**, a **counsellor (vizier)**, a **chariot**, an **elephant**, **two horsemen** and a **foot soldier**. The set is incomplete (a full set would have thirty-two pieces).

- **Date:** estimates range from the late sixth century to about 760. Most popular accounts say mid-seventh to eighth century. This is the earliest set of chess pieces that specialists broadly accept.
- **Where kept:** the State Museum of Samarkand.
- **Why they matter:** the pieces are figurative, carved in the shape of real military units, and they correspond closely to the pieces later Persian sources describe. The king, vizier, chariot, elephant, horse and soldier is precisely the cast of chaturanga. The style of some of the pieces reflects Indian artistic conventions, which supports the idea of transmission from India eastward and westward along the Silk Road.

An article from *Aramco World* on the Afrasiab finds quotes a story set in the year 762 about a man who dreamt of a battle whose moves resembled Chinese chess, and who woke to find a set of chessmen buried in a well. It is an example of how quickly chess entered the folklore of the lands it reached.

7.3 The Dalverzin-tepe figures (earlier, but disputed)

In 1972, in the ruins of Dalverzin-tepe in Uzbekistan, archaeologists found a small ivory **elephant** and a **bull** (or zebu), dated by their finders to as early as the second century CE. Uzbek writers sometimes present them as the oldest chess pieces in the world. The Russian chess historian **Linder** argued that they belong not to chess but to a **forerunner** of chess. Other scholars say their use as game pieces is simply unknown. The sensible conclusion is that they may be game pieces, they may be part of an earlier, different game, and they cannot yet be counted as chess.

7.4 The Nishapur abstract set (late 8th–9th century)

The Metropolitan Museum of Art excavated at **Nishapur** in eastern Iran in the 1930s and 1940s. In a house datable to the early ninth century, archaeologists found **twelve ivory chess pieces** that were probably made around the end of the eighth century. These are the earliest **abstract** pieces known: instead of carved horses and elephants, they have simple geometric shapes. A piece with two small humps stands for the elephant, a block with a slot for the knight, and so on.

The change from figures to abstract forms is a sign of the game's passage into the Islamic world. Scholars have suggested that it reflects a reluctance to depict living beings. The abstract style then spread widely: later Islamic sets were made of ivory, bone, wood, jet,

glass, ceramic, rock crystal and semiprecious stones. The images displayed above in this conversation include a lapis lazuli elephant piece and Nishapur-type pieces.

7.5 The Mozarab pieces (c. 10th century)

In northern Spain, small ivory pieces preserved at a Mozarabic (Christian under Muslim rule) monastery in Leon, often called the **pieces of Saint Genadio**, may be as old as the beginning of the tenth century. Related pieces are held in the Museu de Lleida in Catalonia and in collections in Kuwait. They show the Islamic abstract style crossing the Pyrenees and entering Christian hands.

7.6 The Venafro pieces (c. 980)

At Venafro in southern Italy archaeologists found chess pieces of bone topped with ivory, dated to about 980 CE, now in the Archaeological Museum of Naples. They are among the oldest known European pieces.

7.7 The "Charlemagne" chessmen (c. 1100)

A massive set of carved elephant-ivory chessmen in the Cabinet des Médailles of the Bibliotheque nationale de France in Paris is famous as the **Charlemagne chess set**. It is called that because for centuries it was kept at the abbey of Saint-Denis and a legend tied it to the emperor. Charlemagne lived about three centuries before the set was carved and probably never played chess. The pieces are dated to around 1100 and were probably made in southern Italy, perhaps in Salerno. They are rich in detail, with elephants carrying castles on their backs, riders, and a seated king and queen.

7.8 The Lewis chessmen (12th century)

The best-loved ancient chess pieces in the English-speaking world are the **Lewis chessmen**.

- **Found:** in 1831, in an underground chamber near Uig Bay on the west coast of the Isle of Lewis in the Outer Hebrides, Scotland.
- **Number:** roughly ninety pieces from about four sets, including kings, queens, bishops, knights, rooks and pawns. Most are in the British Museum in London and a smaller group is in the National Museum of Scotland in Edinburgh.

- **Material and origin:** carved mainly from walrus ivory in the twelfth century. They are thought to have been made in Norway (the Hebrides were then under Norwegian rule), perhaps in Trondheim, and may have been part of a merchant's stock being carried from Norway to Ireland.
- **Why they are famous:** the faces are full of character. The queens sit with a hand pressed to the cheek, the knights ride small horses, and the rooks are fierce warriors biting their shields, the "berserkers" of Norse tradition. The bishops are among the oldest known bishops in chess, a sign of how Europe had reshaped the old elephant piece into a figure of its own world.

7.9 Disputed claims for older pieces

Popular lists of "oldest chess pieces" often include items that should be treated cautiously:

- A small ivory piece found in 2002 at **Butrint** in Albania, dated to about 465 CE. If accepted as a chess piece it would push the date of chess back a century, but most specialists are not convinced.
- A piece with a cross on top found in a Byzantine or Roman palace, which would be older still if it really were a chess piece.
- A sandstone figurine found in Jordan in 1991 and dated to the eighth century.
- The elephant and bull from Dalverzin-tepe, discussed above.

Until these are confirmed, the Afrasiab set remains the oldest generally accepted set.

7.10 Where to look for historic pictures

This report cannot embed copyrighted museum photographs, but the following are the images worth searching for, with the institution that holds or publishes each:

Image	Why look at it	Where it is
Afrasiab ivory chessmen	Oldest accepted pieces	State Museum of Samarkand; ChessBase article "The oldest chess pieces in the world"
Nishapur abstract pieces and lapis lazuli elephant	First abstract Islamic pieces	Metropolitan Museum of Art, New York

Image	Why look at it	Where it is
Charlemagne chessmen	Early European carved set	Cabinet des Médailles, Paris
Lewis chessmen	Most famous medieval pieces	British Museum; National Museum of Scotland
Krishna and Radha playing chaturanga	Indian painting of the game on an 8×8 board	Reproduced on World History Encyclopedia
Shahnameh miniatures of the Indian envoy at Khosrow's court	The arrival of chess in Persia	Illustrated Shahnameh manuscripts (mostly from the 14th century onward)
Persian miniature of shatranj players	How the game was played in 1430	Bayasanghori Shahnameh (1430)
Libro de los juegos (Alfonso X, 1283)	Medieval Spanish picture book of chess, dice and tables	Monastery library of El Escorial
Lepakshi relief of Shakuni and Yudhishthira	The dice game of the Mahabharata	Lepakshi temple, Andhra Pradesh

Chapter 8. Persia: from *chaturanga* to *chatrang*

8.1 The route west

The game crossed from India into the **Sasanian Empire** (224–651) in the sixth century or soon after. Persian sources are unanimous that it came from India, and the Persians kept the Indian name, shortened to *chatrang*. The usual guess is that merchants, diplomats and travelers carried it, and that the king mentioned in the Persian texts, **Khosrow I Anushirvan** (reigned 531–579), is the Persian ruler at whose court the game became known.

8.2 The story of Dewasarm and Bozorgmehr

The Pahlavi text *Wizarishn i Chatrang ud Nihishn i Nev-Ardashir* ("The Explanation of Chess and the Arrangement of Backgammon") tells the most famous story of the game's arrival. It is a legend, but a rich one. Its outline is this:

1. In the reign of Khosrow, a great king of India, called **Dewasarm** (also rendered Divsaram or Devisharm), wanted to test the wisdom and knowledge of the Iranians. He set up the game of *chatrang* with a board and pieces and sent it to Khosrow's court with an ambassador, bringing great gifts.
2. The challenge was this: if the Iranian sages could work out the rules and the meaning of the game, India would pay tribute to Persia. If they could not, Persia would owe tribute to India.
3. Khosrow asked for time. In the version told by later Persian writers he asked for **forty days**. His wise men tried and failed, until, on the last day, the youngest and wisest of the king's ministers, **Bozorgmehr** (Wuzurgmihr), stepped forward and explained the game. He is said to have played the Indian envoy and won three times.
4. Bozorgmehr then answered with a challenge of his own: he invented **backgammon** (*nard*, in the text called after Ardashir), and sent it to India. The Indian sages could not decipher it, and so the honor of the contest went to Persia.

The text gives a symbolic account of the new game. According to the *Wizarishn*, Bozorgmehr built the backgammon board like the earth, with thirty counters like the thirty days and nights of the month, fifteen white like the day and fifteen black like the night, and dice like the revolution of the stars. This is cosmology more than history, but it shows how the Persians read their games as models of the universe.

In the tradition retold by the later poet **Ferdowsi** in the *Shahnameh* (about 1010), the Indian gift is a chess set whose pieces are made of **emeralds and rubies** (green against red), and the story includes a framing legend about an Indian king's family quarrel and a mother's grief, in which the game is shown to the queen to explain how her son had died. The details change from version to version, and no Indian source records the Dewasarm story. Scholars generally think the Indian king is probably fictional, or at best an echo of a historical ruler from northern India, such as the Maukhari king of Kannauj, whose embassy may lie behind the legend.

Two lessons come from this tale. First, the Persians did not claim to have invented chess; they said it came from India. Second, the game's arrival was remembered as a **contest of minds between civilizations**.

8.3 What the Pahlavi texts tell us about the game

From the *Wizarishn* and related sources, the Persian game appears as follows:

- The pieces were called **shah** (king), **farzin** (counsellor, later *vazir*), **pil** (elephant), **asp** (horse), **rukh** (chariot) and **piyade** (foot soldier). The names show that the Persians translated or adapted the Indian roles.
- The text speaks of a battle between two rulers, with the rooks at the sides and the counsellor beside the king, and describes the game as a likeness of war.
- It is a two-player game without dice.

8.4 The chess of the heroes

Persian literature also treats chess as a mark of a noble education. The *Karnamak* lists skill at chatrang among the accomplishments of the first Sasanian emperor, Ardashir I, and the *Khusraw ud Redag* has a boy say that he surpasses his companions at chess. As noted in Chapter 3, these texts probably contain later additions, but they show that by the end of the Sasanian period chess was a standard attribute of the cultured Iranian.

8.5 The Persian claim to chess

A minority tradition, found mainly in modern Iranian writing, argues that chess may have had a Persian origin or that the Indian story is overstated. The argument rests on the facts that the earliest *pieces* are Central Asian and Persianate (Afrasiab was a Sogdian and Persianate city, under Islamic rule from 712), and that no early pieces have been found in India. The majority view has not changed: the language of the game is Indian, the Persian texts say it came from India, and the earliest literary mentions are Indian. A neutral way to put it is that chess may have crystallized in the borderlands where Indian and Iranian worlds met, and that the precise place of the first board remains uncertain. What is certain is that the Persians gave chess its second great home.

Chapter 9. The Islamic world: *shatranj* and the first chess literature

9.1 From Persia to the caliphate

When the Arabs conquered Sasanian Persia in the seventh century, they took over its culture, including chess. *Chatrang* became **shatranj** in Arabic, and the pieces mostly kept their Persian names, now Arabized. A Persian folk etymology, charming but false, split the word into *shat* ("hundred") and *ranj* ("worries"), a game of a hundred troubles.

The Arabic names of the pieces:

Arabic / Persian	Meaning	Later European descendant
shah	king	King (from the same word; also "check")
firz or **farzin** (also *wazir*)	counsellor	Queen
fil (with the article, *al-fil*)	elephant	Bishop (Spanish *alfil*, Italian *alfiere*)
faras (or *asb*)	horse	Knight
rukh	chariot	Rook (via Italian *rocco*)
baidaq (or *piyade*)	foot soldier	Pawn

The words *check* and *checkmate* come from Persian **shah** and **shah mat** ("the king is helpless" or, in a popular but disputed derivation, "the king is dead").

9.2 The golden age of shatranj (about 800–1000)

The ninth and tenth centuries produced the first serious **chess literature** in history. The tenth-century bibliography called the *Fihrist* of **Ibn al-Nadim** (compiled in 988) has a section on chess that lists:

- **al-Adli**, *Kitab ash-shatranj* ("Book of Chess"), among the earliest masters and authors;
- **ar-Razi**, *Latif fi'sh-shatranj* ("Elegance in Chess");
- **as-Suli**, *Kitab ash-shatranj* in two volumes;
- **al-Lajlaj**, a book of chess positions or problems (*mansubat*);
- **Aluqlidisi**, a collection of chess problems.

The great players of the age became celebrities. **Abu Bakr as-Suli** (died about 946) was so famous at the game that an Arabic proverb said of a very good player, "He plays chess like as-Suli," and the thirteenth-century biographer Ibn Khallikan used his entry as the occasion to tell the story of chess's invention. Players were ranked in grades, with the highest rank held by the very strongest, and the best could play without seeing the board. Blindfold play is attested in early Arabic sources.

The Arabic books recorded, for the first time in history:

- **opening analysis** (*tabiyat*), lines of opening play that a student could learn;
- **chess problems** (*mansubat*), composed positions in which White wins or draws;
- **the knight's tour**;
- **rules of the game**, including stalemate and the bare king.

This was the beginning of chess as a field of study.

9.3 The caliphs and the courts

Tradition connects several Abbasid caliphs, including **Harun al-Rashid**, with chess, and the game was a favorite of the courts of Baghdad. Courtiers and scholars gambled on it, wrote about it and used it to display wit. The great collections of Islamic art show elaborate sets in ivory, rock crystal, glass and ceramics, which tell us how prestigious the game was.

9.4 Religious debate

Islamic lawyers disagreed about whether chess was permitted. Some objected because of its association with gambling and the figures of living creatures on early sets, and others because it could make people neglect prayer or waste time. Others allowed it as a harmless intellectual pastime as long as no wager was made. The shift to abstract pieces probably reflects part of this debate.

9.5 Big boards and variants

The Islamic world also produced variants of the game. The most famous is the **Tamerlane chess** (*shatranj kamil*, "perfect chess") of the fourteenth century, played on a larger board of ten by eleven squares with extra pieces, associated with the conqueror Timur. Variants of this kind show that the game was now a living tradition that people felt free to change.

9.6 Into Africa, Spain and Sicily

With the spread of Islam, shatranj moved westward across North Africa to al-Andalus (Muslim Spain) and to Sicily, and eastward through Central Asia. It was in Spain and Italy that it came into contact with Christian Europe. The Spanish word **ajedrez** and the Portuguese **xadrez** are still the Arabic *ash-shatranj* in disguise, a memory of this transmission.

Chapter 10. The eastern branches: Southeast Asia, China, Korea, Japan

10.1 Two roads out of India

Murray summarized the spread of the game this way: from India it went west through Persia into the Islamic world and Europe, and east by a second route, probably the same one that carried Buddhism, into Southeast Asia and China. Along the eastern roads the game kept its Indian character in some places and changed radically in others. The prevailing view among historians is that chaturanga is the common ancestor of chess, Chinese *xiangqi*, Korean *janggi*, Japanese *shogi*, Burmese *sittuyin*, Thai *makruk* and Cambodian *ouk chatrang*. The Silk Road and the maritime routes explain how a game could pass between so many cultures.

10.2 Southeast Asia: the closest cousins

The Southeast Asian games are the most faithful descendants of the Indian original.

- **Sittuyin (Myanmar).** The name means "the army of four limbs," since in Burmese *sit* means army, and the word echoes the four-part army of chaturanga. In sittuyin the players arrange most of their pieces on their own half of the board before play begins, and the bishop-like piece moves like the "silver general" of shogi.
- **Makruk (Thailand).** The game of the Thai kingdom retains the slow, old-fashioned shape of shatranj. The minister (*met*) moves one square diagonally, and the elephant-like piece (*khon*) moves one square diagonally or one square forward, the same as the silver-general move that many historians believe was one early Indian elephant move. Pawns promote on the sixth rank. The game is still played in Thai parks and shops.
- **Ouk chatrang (Cambodia).** The name plainly comes from *chaturanga*, and the game is another close cousin of makruk.

- **Catur (Malay and Indonesian).** The very word *catur* is the Sanskrit *chatur*.
- **Co ngu oi (Vietnam).** Vietnam also has a colorful tradition of "human chess" in which people play the pieces.

The elephant piece illustrates how differently the same ancestor developed. Chess historians point to two main versions of the elephant's move: a leap of two squares diagonally (which became the Persian and Arabic *fil*, and then the European bishop through various changes) and a short step of one square diagonally or forward (the silver general), which went to Burma and Siam and later influenced shogi. The version using the leap of two squares, the *alfil*, eventually disappeared from play.

10.3 Tibet and Mongolia

The Tibetan name for the game, *chandaraki*, is plainly derived from *chaturanga*, and the Mongolian *shatar* is probably from the same root. Murray took the names *sittuyin*, the Annamese *chhoen trang* and *chandaraki* as certain reproductions of the Sanskrit, and the Malay *chator* and Mongolian *shatara* as probable ones.

10.4 China: xiangqi

Chinese chess, **xiangqi** ("elephant game"), looks different from chaturanga. Its pieces are flat discs marked with characters, and they stand on the **intersections** of the lines of a board, not in the squares. A river divides the two halves, and a palace marks the area in which the general and his advisors must stay. Its pieces are the **general**, two **advisors**, two **elephants** (which may not cross the river), two **horses**, two **chariots**, two **cannons** and five **soldiers**.

The relationship to chaturanga is debated:

- **The mainstream view** is that xiangqi descends from chaturanga, carried to China along the Silk Road. The chariot, horse, elephant and soldier correspond to the four limbs of the Indian army, and the move of the Chinese elephant (two squares diagonally, but unable to jump) resembles the Indian elephant's move.
- **The minority view**, put forward by the Chinese-American writer **David H. Li** in *The Genealogy of Chess* (1998), argues that the origin of chess lay in China and that the game spread westwards. Li's idea has not been widely accepted, and it remains a minority position.

- China had **older board games**, such as *liubo*, which existed for centuries before chess and which also used pieces on a board. Chess historians study whether these influenced the Chinese game.
- A Tang dynasty story, reported in the *Aramco World* article cited earlier, tells of a man in the year 762 who dreamt of a battle whose moves matched Chinese chess and who woke to find a set of chessmen buried in a nearby well. It shows the game was known in China by the eighth or ninth century at the latest.

10.5 Korea and Japan

Janggi in Korea is closely related to xiangqi. Its elephant move was changed into a slightly longer-reaching version of the horse's move.

Shogi in Japan is the most dramatically transformed relative. Its pieces are wedge-shaped, with the names on the face, and it has a rule that no other chess-family game shares: a captured piece may be brought back into play on the capturer's side (the *drop* rule), introduced in later centuries. The silver general and gold general of shogi are recognizable relatives of the Indian elephant and the minister.

Chess historians sometimes place all of these games on a genealogical "tree," with chaturanga at the root and branches running to Persia and the West, to Southeast Asia, and to East Asia. The details of the branching are still being studied.

Chapter 11. Europe: from *shatranj* to the Queen

11.1 Arrival

Chess reached Europe through several doors at once: **Muslim Spain** (al-Andalus), **Sicily and southern Italy**, and **Byzantium**, where the Greek name *zatrikion* shows the Persian *chatrang*. It also entered Russia from the east. By about the year 1000 it was known in the courts and monasteries of western Europe. The oldest European pieces (Venafro, the Mozarab pieces) date from the tenth century.

As the game moved, it was adapted to Christian Europe's own picture of society:

- The Persian **shah** (king) stayed a king, and his name gave Europe the word for the game itself (Old French *eschecs*, English *chess*) and for *check*.

- The **firz or farzin** (counsellor) became the **queen** in many languages, partly because the Arabic word sounded like a Romance word and partly because Europeans liked to imagine the royal pair.
- The **fil** (elephant) became the **bishop** in English, the *alfiere* (standard-bearer) in Italian, the *alfil* in Spanish and the *fou* (jester) in French. Europeans who had never seen an elephant reinterpreted the odd, two-horned shape of the piece. In Russia the old name for elephant survived as *slon*.
- The **rukh** (chariot) became the **rook** (a tower in several languages).
- The **asp** (horse) became the **knight**.
- The **piyade** (foot soldier) became the **pawn** (from a word for "foot soldier").

11.2 The Church and the game

Chess found a place in monasteries and courts, and also drew criticism. The eleventh-century reformer **Peter Damian** is famous for rebuking a bishop for playing chess. Church authorities worried about gambling and time-wasting. Despite this, chess became a popular activity of nobles and clergy.

11.3 Chess as a mirror of society

Medieval Europe loved moralized chess. About 1300 the Dominican friar **Jacobus de Cessolis** wrote a book that used the pieces as a picture of the social orders: the king, the queen, the bishops, the knights, and the pawns as the various trades and workers. It was one of the most copied and translated books of the age, and in 1474 the English printer **William Caxton** produced an English version, *The Game and Playe of the Chesse*, one of the very first books printed in English.

The great illustrated chess manuscript of the age is the *Libro de los juegos* ("Book of Games"), commissioned by King **Alfonso X of Castile** in 1283, which includes chess problems and charming pictures of Christians and Muslims playing together.

11.4 The revolution of about 1475: the Queen and the Bishop

For roughly seven centuries Europe played the old Arabic game, with a weak counsellor (called the queen) and a short-leaping elephant (the bishop). Then, in the later fifteenth century, the rules changed so quickly that the new game was called *scacchi alla rabiosa*, "mad queen chess." The new rules gave the **queen** the combined powers of rook and

bishop and gave the **bishop** a long diagonal range. Pawns gained the optional two-square first move, with the **en passant** rule following, and castling took its modern shape in the following period.

The oldest surviving text that shows the new game is the Catalan poem **Scachs d'amor** ("Chess of Love"), composed in **Valencia** around **1475** by three poets, **Bernat Fenollar, Narcís Vinyoles and Francí de Castellví**. It is an allegorical poem about a chess game between Mars and Venus' champions, in which the queen (*dama*) and bishop already move as they do today. The exact date is a matter of scholarly discussion, but the poem is widely accepted as the earliest text to show the modern rules. It was followed by the printed books that spread them:

- **Francesc Vicent**, *Llibre dels jochs partits dels schachs* (Valencia, 1495), now lost, the first printed treatise on the new chess;
- **Luis Ramírez de Lucena**, *Repetición de amores y arte de ajedrez* (Salamanca, about 1497);
- **Pedro Damiano**, a Portuguese author, whose book appeared in Rome in 1512.

A popular tale credits **Queen Isabella of Castile** with making the queen powerful, as a tribute to her own rule. Historians doubt the story. The queen's new powers arose in the literature and playing circles of the Spanish and Italian worlds, and Valencia is now celebrated as the cradle of modern chess.

The new queen made chess faster and more tactical. It is the greatest single change in the game since it left India, and it is the reason the game has seemed endlessly rich ever since.

11.5 The first modern masters

After the new rules, European players began to study the game scientifically:

- **Ruy López de Segura**, a Spanish priest, published a book of analysis in 1561, with an opening that still bears his name.
- The **Italian school** of the sixteenth and seventeenth centuries (Polerio, Greco and others) developed brilliant attacking play.
- **François-André Philidor**, a French composer and chess master, published in 1749 a famous book in which he argued that the pawns are the soul of chess, laying the foundations of positional play.

Chapter 12. The modern era

12.1 Standardization

The rules were fixed gradually. **Stalemate** was settled as a draw in the early nineteenth century, replacing the old rule that made it a win or a loss. Clubs, journals and books multiplied, and city teams played each other by correspondence. In 1849 the London makers **Jaques** brought out the **Staunton pattern** of pieces, named for the leading English player **Howard Staunton**, and it was offered to the public on 29 September 1849. Its clear, balanced design became the standard that all international events use today.

Staunton also helped organize the first international tournament, held in **London in 1851** with sixteen leading players from across Europe, which was won by **Adolf Anderssen**.

12.2 Champions and federations

- **1886:** Wilhelm Steinitz is recognized as the first official world champion.
- **1924:** the world chess federation, **FIDE** (Fédération Internationale des Échecs), is founded in Paris.
- The twentieth century saw the Soviet school of chess, the 1972 world championship match of Bobby Fischer against Boris Spassky, and the long reign of Garry Kasparov.

12.3 Machines

In 1997 IBM's **Deep Blue** defeated Kasparov in a match, a landmark in artificial intelligence. In 2017 DeepMind's **AlphaZero** taught itself chess from the rules alone and defeated the strongest conventional engines, with a style that human players found strikingly original. Today the strongest programs run on phones, and online chess brings millions of players together.

12.4 The return to India

The country that gave the world chaturanga took a long time to become a chess power, but it has.

- **1951:** the **All India Chess Federation** is formed.
- **1956:** India makes its debut at the Chess Olympiad in Moscow.

- **1988: Viswanathan Anand** becomes India's first grandmaster, and later wins the world championship five times.
- **2022:** India hosts its first Chess Olympiad, at Chennai, and its teams win bronze medals in the Open and Women's sections.
- **2024:** India wins the gold medals in both the Open and Women's sections at the Budapest Olympiad, and **D. Gukesh** becomes the world champion at the age of eighteen, the youngest in history.
- As of December 2024, India had 85 grandmasters.

It is a satisfying circle: the game of the "four limbs" has come home as one of the strongest chess nations on earth.

Chapter 13. The names of chess and its pieces

13.1 The name of the game in many languages

Language	Name	Note
Sanskrit	*chaturanga*	"Four-limbed"; army and game
Tamil	*chaturangam*	Used in the Shiva legend
Hindi / Urdu	*shatranj*	Via Persian and Arabic
Middle Persian	*chatrang*	Pahlavi texts
Persian / Arabic	*shatranj*	Arabized form
Spanish	*ajedrez*	From *ash-shatranj*
Portuguese	*xadrez*	Same source
Greek (medieval)	*zatrikion*	From Persian
Italian	*scacchi*	From *shah*
French	*échecs*	From *shah*
English	*chess*	From Old French *eschecs*
German	*Schach*	From *shah*
Russian	*shakhmaty*	From *shah*

Language	Name	Note
Turkish	*satranç*	From *shatranj*
Burmese	*sittuyin*	"Army"
Thai	*makruk*	Different word, same family
Cambodian	*ouk chatrang*	From *chaturanga*
Malay / Indonesian	*catur*	From *chatur*
Tibetan	*chandaraki*	From *chaturanga*
Mongolian	*shatar*	Probably from *chaturanga*
Chinese	*xiangqi*	"Elephant game"
Korean	*janggi*	Related to xiangqi
Japanese	*shogi*	"General's game"

The pattern is striking. Two big families of names spread out from the same root: one from the Sanskrit *chaturanga* (Persian *chatrang*, Arabic *shatranj*, Spanish *ajedrez*, Malay *catur*) and one from the Persian *shah*, "king" (Italian *scacchi*, French *échecs*, German *Schach*, English *chess*, Russian *shakhmaty*).

13.2 The pieces through the ages

Role	Sanskrit (chaturanga)	Persian	Arabic	Medieval Europe	Modern English
Ruler	*raja*	*shah*	*shah*	king	King
Counsellor	*mantri*	*farzin*	*firz*, *wazir*	*fers*, *dame*	Queen
Elephant	*gaja*	*pil*	*fil*	*alfin*, *alfil*, *fou*	Bishop
Horse	*ashva*	*asp*	*faras*	*chevalier*	Knight
Chariot	*ratha*	*rukh*	*rukh*	*roc*, *tour*, *rocco*	Rook
Foot soldier	*padati*, *bhata*	*piyade*	*baidaq*	*pedon*, *peon*	Pawn

---

Chapter 14. The legend of the wheat on the chessboard

14.1 The story

The most famous chess story of all is a mathematical parable. It is told in many versions, and this is the usual one.

A wise man named **Sissa ibn Dahir** invented a new game for an Indian king (named Balhait, Shahram, Shirham or Ladava in different tellings). Perhaps his aim was to teach the king, who was proud and careless, that a ruler cannot survive without his people. The king was delighted by the game and offered Sissa any reward he wished.

Sissa's request seemed modest. He asked for one grain of wheat on the first square of the board, two on the second, four on the third, and so on, doubling on every square until the sixty-fourth. The king agreed, almost insulted at the smallness of the request. But when the royal treasurers began to count, they found that the whole kingdom's harvest could not pay the debt. The king saw that the inventor's cleverness in asking was even greater than his cleverness in inventing the game. In some tellings the king rewards him anyway, and in a grim humorous one the king solves his problem by beheading the mathematician.

14.2 The mathematics

The number of grains needed is the sum of 1 + 2 + 4 + … + 2^63, which equals 2^64 − 1, or **18,446,744,073,709,551,615 grains**, far beyond any harvest in history. The thirteenth-century biographer Ibn Khallikan, who gives the first known written account of the story (in 1256), admits that he consulted accountants in Alexandria in order to grasp the size of the number. They told him to imagine the heap reached at the sixteenth square, 32,768 grains, as a certain measure, and to keep multiplying.

The story has been used for centuries to teach **exponential growth**. Modern writers, including Carl Sagan and the authors of *The Limits to Growth*, have used the doubling chessboard to explain the dangers of unlimited growth.

14.3 History or legend?

It is legend. Sissa is regarded as a mythical Brahmin. Until the nineteenth century the story was one of several theories about the origin of chess, and it is now seen as a myth that grew up in the Islamic world. But there is a hint of something real. An earlier Arabic tradition, recorded by the historian **al-Masudi**, says that shatranj was invented under an Indian king who preferred it to backgammon. The Arabic word for a square of the board, *bayt* ("house"), may support the Indian origin of the calculation. As one modern writer puts it, the story may be fictional but not entirely made up: an early mathematical puzzle connected with the chessboard may have inspired the later legend.

Chapter 15. Open questions and competing theories

After a century of study, the history of chess still has unsolved puzzles. A fair account should end with them.

1. **Where exactly was chess first played?** The majority view says northern India. Some argue for the borderlands between India and Iran, and a small minority for China. The earliest physical pieces come from Central Asia, the earliest clear Sanskrit mention from India, and the fullest early stories from Persia.
2. **When?** The cluster of evidence around 600 CE points to the sixth century, but a precursor game might be older. The Dalverzin-tepe figures and the Butrint piece are tantalizing, but unproven.
3. **Was there a dice phase?** The Cox–Forbes theory said yes. Murray's critique said no. Most scholars now say that two-player, dice-free chaturanga is the original and that chaturaji is a later variant, but the Mahabharata's mention of ivory pieces moved by dice leaves room for older dice-based board games that influenced it.
4. **What was the elephant's original move?** Two squares diagonally with a leap, a short step diagonally and forward, or something else? The Asian descendants preserve different answers.
5. **What were the exact rules?** No Indian rulebook of the early period survives, so the rules are reconstructed from Persian and Arabic sources and later descendants.
6. **Why so few early Indian pieces?** Perhaps because they were made of perishable materials, perhaps because archaeology of this question has barely begun.

7. **How are the Chinese games related?** Descent from chaturanga, independent development, or a shared parent game are all still argued.

Glossary

- **Akshauhini:** the largest division of the epic army, with equal numbers of chariots and elephants, three times as many horses and five times as many foot soldiers.
- **Alfil:** the old elephant piece, which leaps two squares diagonally.
- **Ashtapada:** the 8×8 uncolored board used for chaturanga and for dice race games.
- **Chatrang:** the Middle Persian form of the game.
- **Chaturaji:** the four-handed, dice-driven Indian game described by al-Biruni.
- **Chaturanga:** "four limbs"; the four-part army and the ancient Indian game.
- **Farzin / Ferz:** the Persian counsellor, ancestor of the queen.
- **Gaja:** Sanskrit for elephant.
- **Mansubat:** Arabic chess problems.
- **Mantri:** the minister piece.
- **Padati / Bhata:** the foot soldier.
- **Ratha:** the chariot.
- **Shah mat:** Persian phrase behind "checkmate."
- **Shatranj:** the Arabic and Persian name for the old game.
- **Vyuha:** a battle formation in the Mahabharata.

Sources and further reading

Book-length scholarship:

- H. J. R. Murray, *A History of Chess* (Oxford, 1913), the foundation of the field.
- Henry A. Davidson, *A Short History of Chess* (1949).
- Harry Golombek, *Chess: A History* (1976).
- Richard Eales, *Chess: The History of a Game* (1985).
- David H. Li, *The Genealogy of Chess* (1998), for the minority Chinese-origin argument.
- John Gollon, *Chess Variations: Ancient, Regional, and Modern* (1968).

Online sources consulted for this report:

- Encyclopaedia Britannica, entries on chess and chaturanga.
- Wikipedia, articles on Chaturanga, Chaturaji, Shatranj, Chatrang, Alfil, Shakuni, Sissa (mythical brahmin), the wheat and chessboard problem, Scachs d'amor, and Chess in India.
- Encyclopaedia Iranica, entry "Wizarishn i Chatrang ud Nihishn i Nev-Ardashir."
- ChessBase articles: "The oldest chess pieces in the world" (André Schulz); "On the origins of chess" (Sergio Ernesto Negri); "The millennial journey of chess along the Silk Road"; the series on *Scachs d'amor*.
- Chess Variants Pages, "Chaturanga" (comparing Murray, Golombek, Davidson and Gollon).
- World History Encyclopedia, "The History of Chess" (Howard Burton).
- *Aramco World*, "The Game of Kings" (2009).
- UNESCO Silk Roads, "Traditional Strategy Games along the Silk Roads: Chess."
- Edward Winter's Chess Notes and chesshistory.com, for the discussion of the Lothal claim.
- Burlington Magazine review of the Al-Sabah Collection catalogue on chess and backgammon pieces from Islamic lands.
- KWABC (Valencia) essays by José Garzón on the cradle of modern chess.
- Kolachana and Wujastyk, "Gameplay as Foreplay at a Medieval Indian Court" (2022), on the *Manasollasa*.
- Popular and textbook sources for the Mandodari, Ganesha and Shiva legends (Organiser, Outlook India, RBSE Class 10 textbook) and the review of Aashish Kaul's novel *The Queen's Play*.

A note on limits

This report was assembled from the sources above and from standard scholarship. Dates for the oldest pieces, the details of early rules and the content of origin legends vary between sources, and where they do the report says so. Some details in the Persian and Arabic stories are reported as they appear in the traditions, not as verified history. For an assignment, cite the primary scholarly works (Murray, Eales, Golombek) and the original articles rather than this summary.